MSR training dataset

Contents

Introduction

This exercise provides a hands-on practical introduction to working with the Model Supply Regions (MSR) framework. The MSR framework identifies attractive sites for variable renewable energy (VRE) deployment that can serve as candidate investment options in energy-system models.

In this fictional example, consultants are preparing a national power system capacity expansion study for Ecuador. Their existing model represents the country as a single renewable-energy region, which hides important differences in solar and wind quality, distance from existing road and grid infrastructure, and land availability. They intend to use the MSR framework to construct spatially explicit renewable model supply regions that can be used in the capacity expansion model.

Downloading MSR framework and accompanying software

GitHub repository

The MSR framework is provided as a collection of Python scripts, control files and input datasets compiled within the GitHub repository. Clone the MSR framework GitHub repository, including the training material, to a local folder on your system.

The repository contains the following folders and files:
1_msr_creator/
2_profile_generator/
3_attribute_combiner/
4_screener/
5_clustering/
inputs/
outputs/
envs/
training/
README

Each workflow folder (1-5) contains the relevant Python script and control file. The ***inputs*** folder stores the required input datasets, while the ***outputs*** folder stores the results generated by the MSR framework.

Python environment

Anaconda (https://www.anaconda.com/download/success?reg=skipped) is used to manage the Python environment and the packages required by the MSR framework. The environment is defined in the *msr_env.yaml* file, located in the ***envs*** folder.

👉 Open an Anaconda Prompt or terminal, navigate to the ***envs*** folder and create and activate the environment using the following commands:

```
conda env create -f msr_env.yaml
conda activate msr_env
```

Spyder is used as integrated development environment (IDE) for running the Python scripts (launch *spyder* from the terminal inside the *msr_env* environment).

QGIS

QGIS is used to visualise the spatial input data and the outputs generated by the MSR framework.

Part 1: MSR Creator

The MSR Creator **processes regional geospatial datasets** and converts suitable renewable-resource areas into **Model Supply Regions**.

The workflow consist of five stages:

1. Process regional geospatial inputs.
2. Score land-suitability layers.
3. Identify competitive renewable-resource areas.
4. Polygonise contiguous suitable areas.
5. Attribute the resulting MSRs with capacity, infrastructure-distance, land-cover class, climate class, elevation, and other relevant metrics.

Part 1.1: Regional geospatial inputs

The infrastructure and land characteristics layers are input datasets stored in the ***inputs*** folder, describing Ecuador's existing **road and grid** networks, physical **geography**, and renewable **resource availability**.

When exploring the topographic characteristics of Ecuador (here excluding the Galápagos islands), the country is bisected from north to south by the Andes mountains. This is clearly visible in the elevation and slope layer (**Figure 1**).

Across the mountain range, there are areas with **high solar and wind resource potential** (**Figure 2**).

Divided across both sides of the Andean range there are distinct Köppen-Geiger climate and land-cover classes (**Figure 3**).

Given Ecuador's diverse landscape, it is important to **represent distinct sites** for solar PV and wind, rather than a single renewable energy representation, in order to reveal important **variations in resource strength, land availability, and infrastructure access**.

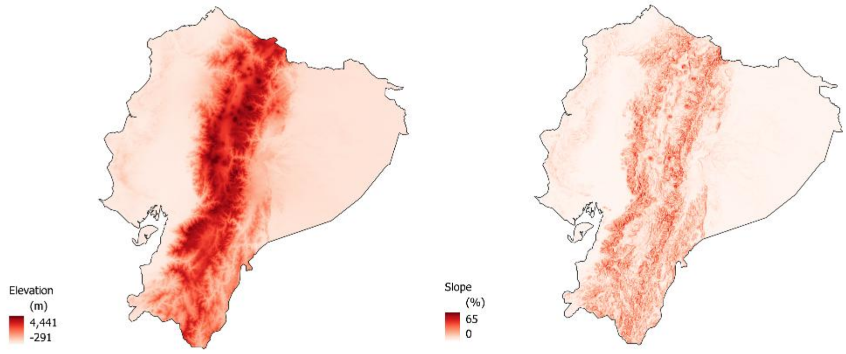

Figure 1: *a) Elevation (meters above sea level, m.a.s.l) in Ecuador from the NASA Shuttle Radar Topography Mission (STRM). b) Slope (%) calculated from the elevation raster. Basemap from the Database of Global Administrative Areas (GADM).*

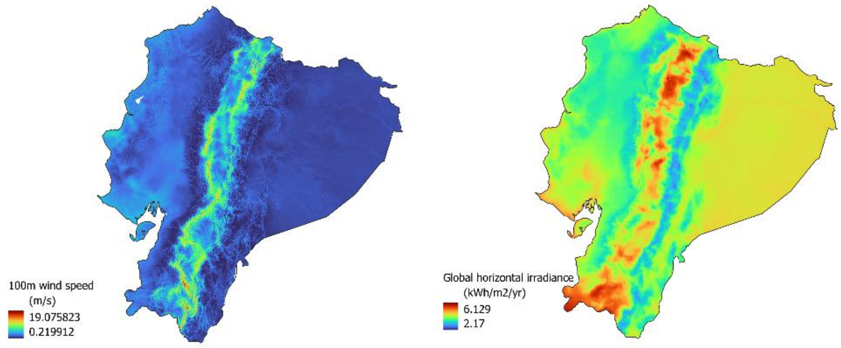

Figure 2: a) Mean wind speed at 100 m (m/s) in Ecuador from the Global Wind Atlas (GWA). b) Global horizontal irradiance (kWh/m²/yr) in Ecuador from Global Solar Atlas (GSA).

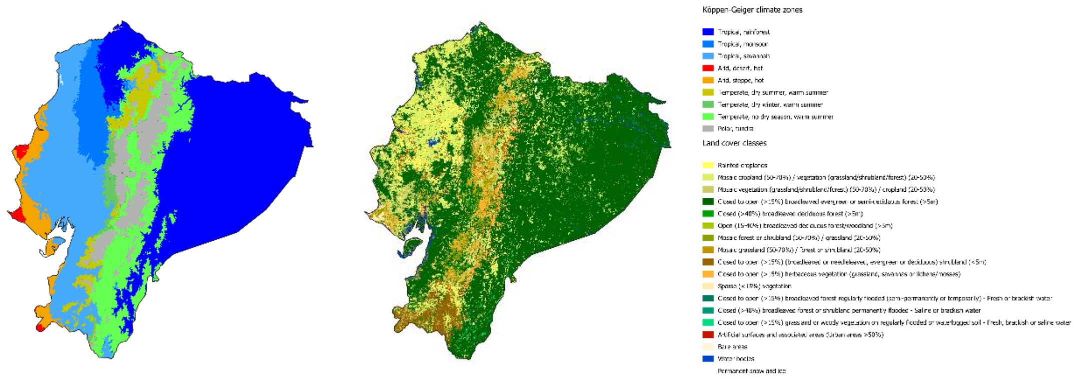

Figure 3: a) Köppen-Geiger climate classes in Ecuador. b) Land-cover classes in Ecuador from ESA GlobCover.

Part 1.2: Exclusion criteria

The MSR Creator **excludes areas that are not considered suitable** for renewable-energy development.

The exclusion criteria, which can be strengthened or loosened by the user, include:

- Very **densely populated areas**, e.g. cities;
- Locations above a certain elevation and slope;
- Selected classes of land-cover;
- Natural reserves and other protected sites;
- Sites outside of a certain vicinity of existing road and electricity networks can be discarded;
- Sites where average VRE resources (irradiation and wind speed) are below a certain minimum threshold, typical for commercial exploitation, can be discarded.

The above criteria define the exclusion areas, and only regions that meet all criteria are retained as suitable land for subsequent processing for wind and solar PV deployment (**Figure 4**).

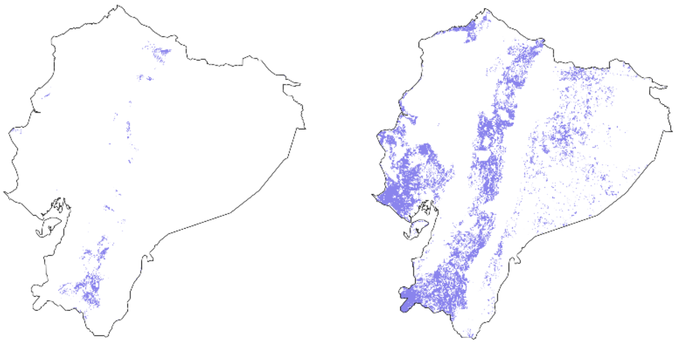

Figure 4: Example of suitable land area (purple) remaining after the application of exclusion criteria for a) wind and b) solar PV in Ecuador.

Part 1.3: Setting up the control file

The control file for the MSR Creator script, ***control_file_msr_creator.xlsx***, is an Excel workbook containing four worksheets:

- *paths*
- *datasets*
- *configurations*
- *parameters*

👉 Open the ***control file*** (MSR creator) and go to the *paths* worksheet.

This worksheet specifies the paths locating the input and the output folders. Specify the paths to the ***inputs*** and ***outputs*** folders in the appropriate cells (see **Figure 5**).

parameter	value	comments
input_folder_datasets	C:\Users\adm_sesterl\OneDrive - Vrije Universiteit Brussel\Documenten\VUB Work Files\26001 MSR tutorial\inputs	Folder path where input datasets are located.
output_folder	C:\Users\adm_sesterl\OneDrive - Vrije Universiteit Brussel\Documenten\VUB Work Files\26001 MSR tutorial\outputs	Folder path where outputs are located.

Figure 5: Entering file paths to input and output folders in the ***control file*** → *paths worksheet.*

👉 Go to the *datasets* worksheet.

This worksheet specifies the file names of the required input datasets in the value column (B) as discussed in **Part 1.1: Regional geospatial inputs**. A description of each dataset is provided in the *comments* column (D).

The file names required for this exercise are listed in ***training_datasets.xlsx*** → *msr_creator* worksheet (the corresponding files are in the ***inputs*** folder). The file names are already arranged in the correct order. **Copy them directly into the value column of the control file** (see **Figure 6**).

	parameter	value	source	comments
2	file_name_regions	region_names		File name with region names (.csv)
3	file_name_region_boundaries	Ecuador		File name of shape file carrying region boundaries
4	file_name_population_density	ECU_lspop	Oak Ridge National Laboratory's LandScan 2019. Resolution: 1x1km2	File name of image carrying population density
5	file_name_urban_area_load_centers	ne_10m_urban_areas_landscan	Natural Earth. Resolution: 10m	File name of image carrying urban areas (for naming purposes; not needed for filtering)
6	file_name_land_cover	ECU_GLOBCOVER	European Space Agency's GlobCover 2009. Resolution 300x300m2	File name of image carrying land cover
7	file_name_elevation	ECU_SRTM	Shuttle Radar Topography Mission (STRM). Resolution 30x30km2	File name of image carrying elevation
8	file_name_climate_classes	ECU_koppen_geiger	Beck et al. (2023). Resolution: 1x1km2	File name of image carrying climate zones
9	file_name_protected_areas	ECU_WDPA	World Database on Protected Areas. Resolution: 300×300m2	File name of image carrying protected areas
10	file_name_water_bodies	ECU_HydroLAKES	HydroLAKES	File name of image carrying water bodies
11	file_name_roads	ECU_GRIP	Global Roads Inventory Project (GRIP)	File name of shapefile carrying road network
12	file_name_substations	ECU_OSM_S	Open street map (OSM)	File name of shapefile carrying substations
13	file_name_power_grid	ECU_gridfinder_allTDgrid	GridFinder 2020	File name of shapefile carrying transmission and distribution grid
14	file_name_transmission_grid	ECU_gridfinder_T	GridFinder 2020	File name of shapefile carrying transmission grid
15	file_name_distribution_grid	ECU_gridfinder_D	GridFinder 2020	File name of shapefile carrying distribution grid
16	file_name_continent_distance_surface_tgrid			*File name of shapefile carrying transmission grid from neighbouring countries (for very rare cases, in case grid distance to be calculated w.r.t. neighbouring grid)*
17	file_name_buffer_distance	buffer_distance		File name with corresponding road and grid buffer distance per region (.csv)
18	file_name_ghi_map	ECU_GHI	Global Horizontal Irradiation (GHI). Global Solar Atlas. Resolution: 1x1km2	File name of image carrying global horizontal irradiation
19	file_name_wind_speed_map	ECU_wind_speed_100m	Wind speed 100m. Global Wind Atlas. Resolution: 250x250m2	File name of image carrying wind speeds
20	file_name_dni_map			File name of image carrying direct normal irradiation
21	file_name_climate_class_legend	koppen_geiger_legend		File name matching climate classes with descriptions (.csv)
22	file_name_land_cover_legend	land_cover_legend		File name matching land cover categories with descriptions (.csv)

paths | datasets | configurations | parameters

*Figure 6: Filling the **control file** → datasets worksheet with input data file names.*

👉 Go to the *configurations* worksheet.

This worksheet specifies the code run configurations and determines which technologies and workflow stages are included. For the first run, enable wind by setting the *run_code_for_wind* configuration equal to 1 and disable the other technologies. Enable *stages 1, 2, 3, 4* and *5*. See **Figure 7**.

	parameter	value	comments
2	run_code_for_solar_pv	0	Options=[0-No, 1-Yes]. Only select one at a time.
3	run_code_for_solar_csp	0	Options=[0-No, 1-Yes]. Only select one at a time.
4	run_code_for_wind	1	Options=[0-No, 1-Yes]. Only select one at a time.
5	run_code_for_offshore_wind	0	Options=[0-No, 1-Yes]. Only select one at a time.
6	perform_stage_1_prepare_input_datasets	1	Options=[0-No, 1-Yes]
7	perform_stage_2_score_input_datasets	1	Options=[0-No, 1-Yes]
8	perform_stage_3_get_resource_potential	1	Options=[0-No, 1-Yes].
9	perform_stage_4_polygonization	1	Options=[0-No, 1-Yes]
10	perform_stage_5_attribution	1	Options=[0-No, 1-Yes]
11	grid_buffered_search	0	Options=[0-No, 1-Yes]. Search in buffered area around grid.
12	roads_buffered_search	0	Options=[0-No, 1-Yes]. Search in buffered area around roads.

paths | datasets | configurations | parameters

*Figure 7: Setting run configurations for the MSR creator using the **control file** → configurations worksheet.*

👉 Go to the *parameters* worksheet.

This worksheet specifies general and technology-specific parameters, including resource thresholds, slope and elevation limits, population thresholds, suitable land-cover classes, minimum and maximum MSR capacities, etc. as discussed in **Part 1.2: Exclusion criteria**.

The parameters required for this exercise are listed in ***training_parameters.xlsx*** → *msr_creator* worksheet. The parameters are arranged in the correct order and can be copied directly into the *value* column of the ***control file***, as shown in **Figure 8**.

	parameters	value	comments
2	**General parameters**		
3	population_threshold	100	Maximum allowed population density; pixels above this threshold are excluded. Units: inhabitants/km².
4	msr_max_capacity_threshold	2700	Maximum deployable capacity of a single MSR; larger MSRs are dividid using a square grided mesh structure. Unit: MW.
5	msr_min_capacity_threshold	20	Minimum deployable capacity of a single MSR; smaller MSRs are excluded. Unit: MW.
6	elevation_threshold	2000	Maximum allowed elevation; pixels above this value are excluded. Unit: m above sea level.
7	land_cover_classes	11, 14, 20, 30, 110,	Suitable land-cover classes.
8	road_type	5	Options= [1,2,3,4,5]. Lowest road hierarchy included in road searches and distance calculations. Values: 1=highway, 2=primary, 3=secondary, 4=tertiary, 5=local. More information on road types: https://iopscience.iop.org/article/10.1088/1748-9326/aabd42
9	band_count_for_multi_resolve_algorithm	5	Options= [1,2,3...]. Number of resource-quality bands used to group and polygonise contiguous suitable pixels.
10			
11	**Solar PV parameters**		
12	pv_ghi_threshold	4	Minimum GHI required for inclusion; pixels below this value are excluded unless threshold relaxation is enabled. Unit: kWh/m²/day.
13	pv_slope_threshold	20	Maximum allowed terrain slope; pixels above this value are excluded. Unit:
14	pv_land_discount_factor	10	Share of technically suitable land assumed to be available for PV deployment. Unit: %.
15	pv_footprint	33	Installed PV capacity per square kilometre of developed land. Unit:
16			
17	**Solar CSP parameters**		
18	csp_dni_threshold	4	Minimum DNI required for inclusion; pixels below this value are excluded unless threshold relaxation is enabled. Unit: kWh/m²/day.
19	csp_slope_threshold	5	Maximum allowed terrain slope; pixels above this value are excluded. Unit:
20	csp_land_discount_factor	10	Share of technically suitable land assumed to be available for CSP deployment. Unit: %.
21	csp_footprint	17	Installed CSP capacity per square kilometre of developed land. Unit:

paths | datasets | configurations | parameters

Figure 8: Entering exclusion parameters in the ***control file*** *→ parameters worksheet.*

Part 1.4: Running the MSR Creator for wind power

Once the input datasets and control file have been prepared, the MSR creator script can be run.

> 🏃 **CODE RUN** 🏃
>
> Navigate to the ***1_msr_creator*** folder and run *msr_creator.py*.

Follow the log messages in the terminal to monitor the progress of the workflow. The script reports the active region, technology and workflow stage. The run is complete when the message "*MSR Creator workflow completed*" appears in the terminal.

In the ***outputs*** folder, the script creates a new folder *1_msr_creator* with subfolders for each modelled region. For this exercise, the outputs are stored in the ***Ecuador*** subfolder, and intermediate outputs are organised by workflow stage.

The region-specific input datasets are stored in the ***stage1_input_datasets*** folder. (The outputs from stage 1 were shown in **Part 1.1: Regional geospatial inputs**.)

The suitability and exclusion layers are stored in the ***stage2_scored_datasets*** folder providing the suitable area according to each separate exclusion layer.

The resource competitive area, combining the different exclusion layers, is located in the ***stage3_competitive_resource_area*** folder.

The suitable MSRs locations are defined in the *wind_final_msrs.shp* shapefile and is stored in the ***stage5_attribution*** folder (**Figure 9**). (The suitable land area for wind and solar PV MSRs were shown in **Part 1.2: Exclusion criteria**.)

The region subfolder also contains summary figures showing the composition of MSR areas by elevation range, land use, and climate zone (**Figure 10**).

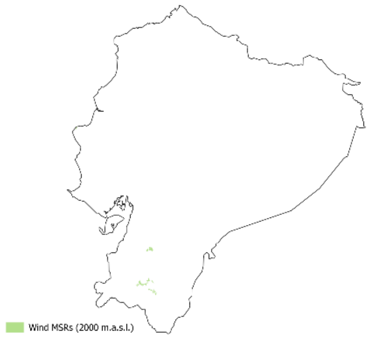

Figure 9: *Example wind MSRs in Ecuador with an elevation threshold at 2,000 m.a.s.l.*

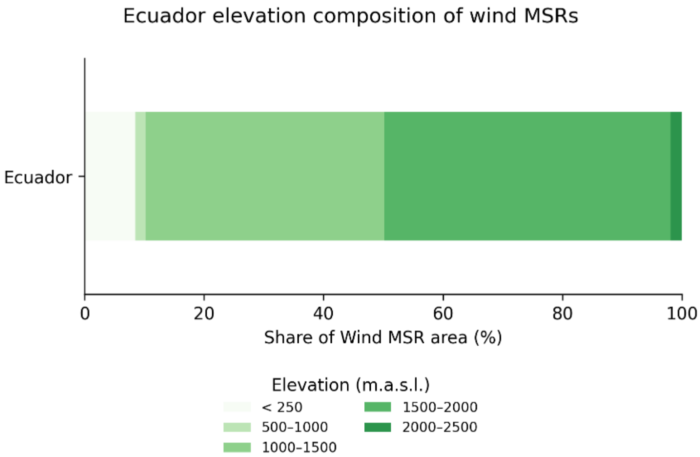

Figure 10: *Example composition of wind MSRs by elevation in Ecuador (for a run with altitude exclusion above 2000 m.a.s.l.).*

Part 1.5: Changing exclusion criteria

Given Ecuador's topography and resource potential, the strongest wind resources are located in the Andes mountain range. In this exercise, the elevation threshold is increased from 2,000 to 3,000 m.a.s.l to investigate how access to higher-elevation areas affects the resulting MSRs.

👉🏼 Go to the ***control file*** → *configurations* worksheet.

Keep the *run_code_for_wind* configuration equal to 1 and leave the other technologies disabled. **Disable** *stage 1*, as the region-specific input datasets were previously created, and leave *stages 2, 3, 4* and *5 enabled*.

👉🏼 Go to the ***control file*** → *parameters* worksheet. Change the *elevation_threshold* parameter from 2,000 to 3,000 m.a.s.l.

🏃 **CODE RUN** 🏃

Run *msr_creator.py* again.

A new shapefile is created in the ***stage5_attribution*** folder, including suitable MSR locations for the adapted elevation threshold (**Figure 11**). The composition of MSR areas by elevation range indicates that the majority of the suitable MSR area is located at higher-altitude areas between 2,000 to 3,000 m.a.s.l. (**Figure 12**).

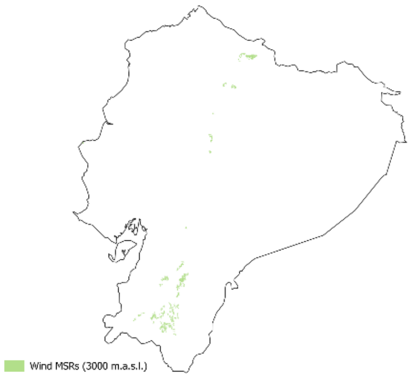

Figure 11: Example wind MSRs in Ecuador with an elevation threshold at 3000 m.a.s.l.

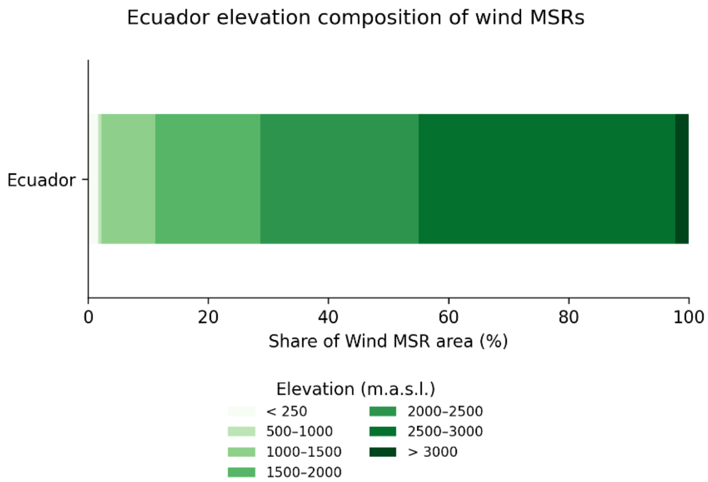

Figure 12: Example composition of wind MSRs by elevation in Ecuador (for a run with altitude exclusion above 3000 m.a.s.l.).

As indicated, the region subfolder also contains summary figures showing the composition of suitable MSR area by land-cover class (example in **Figure 13**) and Köppen-Geiger climate class (example in **Figure 14**).

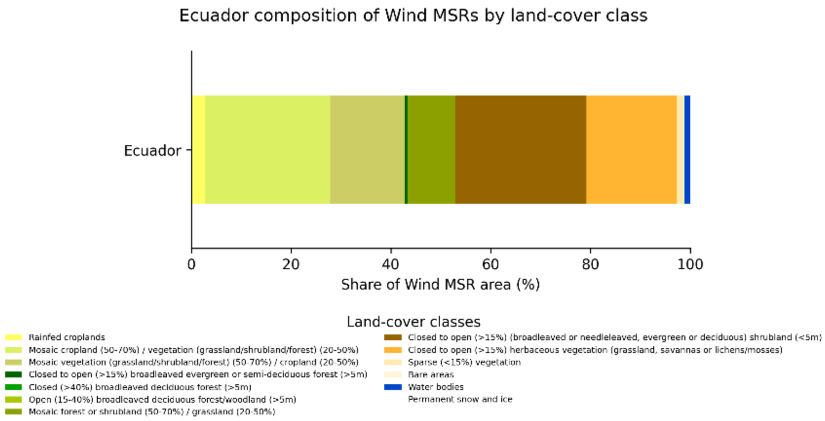

Figure 13: Example composition of wind MSRs by land-cover class in Ecuador.

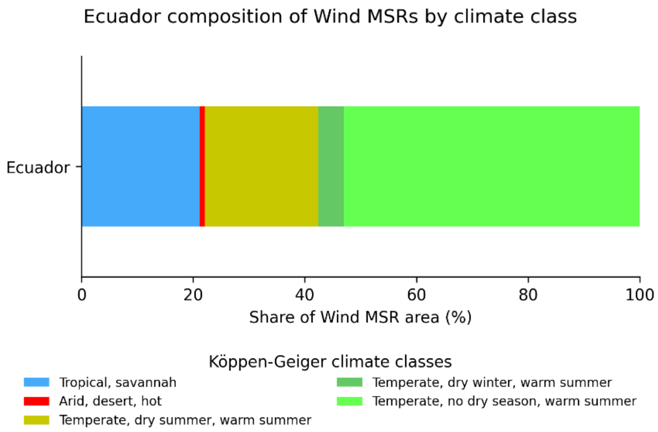

Figure 14*: Example composition of wind MSRs by Köppen-Geiger climate class in Ecuador.*

Part 1.6: Running the MSR Creator for solar PV

Here, we will investigate suitable solar PV MSRs and repeat the MSR Creator workflow.

👉🏼 Go to the ***control file*** → *configurations* worksheet.

👉🏼 Enable solar PV technology by setting the *run_code_for_solarpv* configuration equal to 1 and disable the other technologies. Enable *stages 1*, *2*, *3*, *4* and *5*.

> 🏃 **CODE RUN** 🏃
>
> Run *msr_creator.py* again.

A new shapefile is created in the ***stage5_attribution*** folder including suitable MSR locations for solar PV (**Figure 15**).

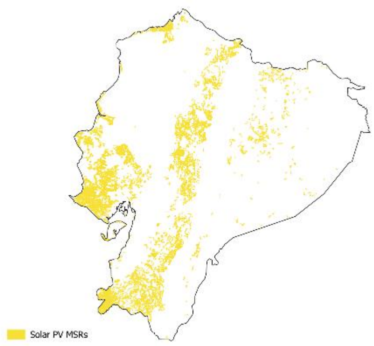

Figure 15*: Example solar PV MSRs in Ecuador.*

Part 2: Profile Generator

The Profile Generator extracts meteorological time series from netCDF *(.nc)* files obtained from the ECMWF's ERA5 reanalysis (https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview), available through the Climate Data Store, and uses it to calculate time series of power generation potential at the created MSR sites.

Part 2.1: Obtaining meteorological input data

In principle, the meteorological input data can be obtained by creating an account on the Climate Data Store (CDS), obtaining a CDS API (more information can be found on https://cds.climate.copernicus.eu/how-to-api), and running a script to download the relevant data.

For the Profile Generator, the data must be downloaded and organised in the following format:

- One file of meteorological data per year (to avoid too large data files)
- Hourly resolution (all hours of the year)
- .netCDF format
- File name: *[year]_[identifier].nc*, for instance "*2012_ssrd.nc*" for *ssrd* data (*surface solar radiation downwards*) for the year 2012

For instance, downloading *ssrd* data for the year 2012 for Ecuador would be done by running the following script after having installed the CDS API:

```
import cdsapi

dataset = "reanalysis-era5-single-levels"
request = {
    "product_type": ["reanalysis"],
    "year": ["2012"],
    "month": ["01", "02", "03", "04", "05", "06", "07", "08", "09", "10", "11", "12"],
    "day": [
        "01", "02", "03", "04", "05", "06", "07", "08", "09", "10", "11", "12",
        "13", "14", "15", "16", "17", "18", "19", "20", "21", "22", "23", "24",
        "25", "26", "27", "28", "29", "30",  "31"
    ],
    "time": [
        "00:00", "01:00", "02:00", "03:00", "04:00", "05:00", "06:00", "07:00", "08:00", "09:00", "10:00", "11:00",
        "12:00", "13:00", "14:00", "15:00", "16:00", "17:00", "18:00", "19:00", "20:00", "21:00", "22:00", "23:00"
    ],
    "data_format": "netcdf",
    "download_format": "unarchived",
    "variable": ["surface_solar_radiation_downwards"],
    "area": [1.5, -81, -5, -75.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

The following meteorological datasets are necessary to run the Profile Generator:

- Surface solar radiation downwards (*ssrd*) *[for solar PV]*
- 2-m temperature (*t2m*) *[for solar PV and wind]*
- 10-m wind speed (*u10, v10*) *[for wind]*
- 100-m wind speed (*u100, v100*) *[for wind]*
- geopotential (*z*) *[for wind]*

For this exercise, the data has already been downloaded and is located in the ***inputs*** folder, in separate .nc files for the years 2012, 2013 and 2014.

👉 Open the ***control file*** (Profile Generator) and go to the *paths* worksheet. Enter here again the correct paths to the ***inputs*** and ***outputs*** folders, as in **Figure 5**.

Part 2.2: Setting up the control file

👉 Go to the *datasets* worksheet. Link here the netCDF files, plus resource rasters and a few data tables in the ***input*** folder, to the correct entries in the table. The *.nc* files will be linked with the *[identifier]* in their file name (see above). The result should look as in **Figure 16**.

	A	B	C
1	datasets	value	comments
2	file_name_resource_raster_solarpv	ECU_GHI	Resource raster naming (.tif file) in *inputs* folder
3	file_name_resource_raster_wind	ECU_wind_speed_100m	Resource raster naming (.tif file) in *inputs* folder
4	10m_component_of_wind	10m	identifier for ERA 5 data .nc files (u10, v10) in input folder. The file name must have a year identifier in front, e.g. *2012_[identifier].nc*
5	100m_component_of_wind	100m	identifier for ERA 5 data .nc files (u100, v100). Idem.
6	2m_temperature	t2m	identifier for ERA 5 data .nc files (t2m). Idem.
7	geopotential	z	identifier for ERA 5 data .nc files (z). Idem.
8	surface_solar_radiation_downwards	ssrd	identifier for ERA 5 data .nc files (ssrd). Idem.
9	file_name_IEC_power_curves	Wind speed IEC Classes	Wind turbine power curves for three IEC classes
10	file_name_regions	region_names	File name of csv file carrying list of regions for which MSR hourly profiles will be created.
11	file_name_region_utc_offset	region_utc_offsets	File name of csv file with offsets per region of time zone compared to UTC
12			

paths | datasets | configurations | parameters | +

Figure 16*: Linking entries to the correct datasets in the* ***control file*** → *datasets worksheet.*

👉 Go to the *configurations* worksheet. Select here *run_code_for_wind* = *1* and set the others to zero, to run the Profile Generator for wind.

👉 Go to the *parameters* worksheet. Set the years to analyse under *weather_years* as semicolon-separated list, as in **Figure 17**. The *reference_year* can be any year in the *weather_years* set (it is necessary to perform the spatial downscaling of wind profiles to higher resolution).

	A	B	C
1	parameters	val	comments
2	weather_years	2012; 2013; 2014	List of weather years. Separate with semicolon, i.e. *2012; 2013; 2014* for the period 2012-14.
3	reference_year	2012	*[only for wind]* The reference year used for fitting linear regression in wind speed bias-correction.
4	wind_hub_height	100	*[only for wind]* Units: meters

Figure 17*: Specifying the Profile Generator parameters in the* ***control file*** → *parameters worksheet.*

Part 2.3: Running the Profile Generator

> 🏃 **CODE RUN** 🏃
>
> Run *profile_generator.py* for wind power.

Follow the log messages in the terminal to monitor the progress of the workflow. The script reports the active region, technology and workflow stage. The run is complete when the message "*Profile Generator workflow completed*" appears in the terminal.

In the ***outputs*** folder, the script creates a folder *2_profile_generator* with a subfolder for each modelled region. The output consist of .csv files with time series for all MSRs in UTC and local time, plus graphics showing statistics of diurnal and seasonal profiles of capacity factors (weighted by MSR size), as in the example in **Figure 18** and **Figure 19** for wind power.

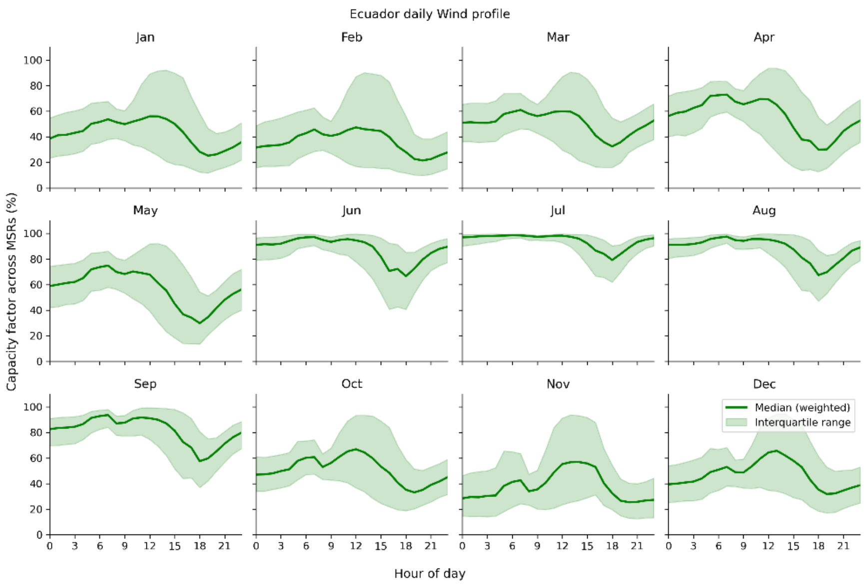

Figure 18: *Example diurnal capacity factor profiles for wind power per month across MSRs after the Profile Generator run.*

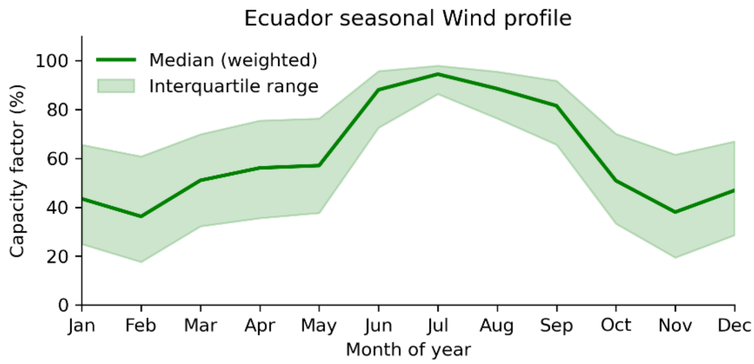

Figure 19: *Example seasonal capacity factor profile for wind power across MSRs after the Profile Generator run.*

Let us now generate the profiles for solar PV.

👉🏼 Go to the *configurations* worksheet. Select here *run_code_for_solar_pv = 1* and the others to zero.

🏃 **CODE RUN** 🏃

Run ***profile_generator.py*** for solar PV. Have a look at the output profiles.

Part 3: Attributor Combiner

The attributor combiner determines, for each individual MSR, the production and cost attributes including loss adjusted annual capacity factors and three part-wise Levelised Cost of Electricity (LCOE) metrics for supply, grid and substation connection, and road connection.

Part 3.1: Setting up the control file

Open the **control file** for the Attributor Combiner script, ***control_file_attributor_combiner.xlsx***, that includes the same worksheet structure as the other control files.

👉 Open the ***control file*** → *paths* worksheet.

Specify the paths to the ***inputs*** and ***outputs*** folders in the appropriate cells, as in **Figure 5**.

👉 Go to the *datasets* worksheet.

The file names required for this exercise are listed in ***training_datasets.xlsx*** → *attributor_combiner* worksheet. The file names are arranged in the correct order. **Copy them directly** into the value column of the **control file**, which should look as in **Figure 20**.

	A	B	C
1	**datasets**	**value**	**comments**
2	file_name_resource_raster_solarpv	ECU_GHI	Resource raster naming (.tif file) in *inputs* folder
3	file_name_resource_raster_wind	ECU_wind_speed_100m	Resource raster naming (.tif file) in *inputs* folder
4	file_name_regions	region_names.csv	Name of region names table (input folder)
5	file_name_cost_assumptions	cost_assumptions.csv	Name of table with cost (CAPEX, OPEX, etc.) assumptions (input folder)
6			

paths | datasets | configurations | parameters

Figure 20: *Specifying the input datasets to run the Attributor Combiner script in the* ***control file*** → *datasets* worksheet.

👉 Go to the *configurations* worksheet.

For the first run, enable wind by setting the *run_code_for_wind* configuration equal to 1 and disable the other technologies. Enable *local_time_profile* to use the local time profiles that were generated by the Profile Generator.

👉 Go to the *parameters* worksheet.

The parameters required for this exercise are listed in ***training_parameters.xlsx***, in the *attributor_combiner* worksheet. The parameters are arranged in the correct order. Copy them directly into the *value* column of the ***control file***.

Part 3.2: Running the Attributor Combiner

👉 Once the control file has been prepared, navigate to the ***3_attributor_combiner*** folder.

🏃 **CODE RUN** 🏃

Run *attributor_combiner.py* for wind power.

Follow the log messages in the terminal to monitor the progress of the workflow. The script reports the active region, technology and workflow stage. The run is complete when the message "*Attributor Combiner workflow completed*" appears in the terminal.

In the ***output*** folder, the script creates a folder *3_attributor_combiner* with a subfolder for each modelled region. Here, the outputs are stored in the ***Ecuador*** subfolder where the attributed wind MSRs are defined in the *wind_prescreen.shp* shapefile. This new shapefile **contains cost attributes for every MSR** (related to power generation, transmission, and road infrastructure), as well as **corresponding LCOE values**.

The region subfolder also contains summary figures showing the composition of suitable MSR areas by distance from the grid (example in **Figure 21**), capacity factor (example in **Figure 22**) and LCOE (example in **Figure 23**).

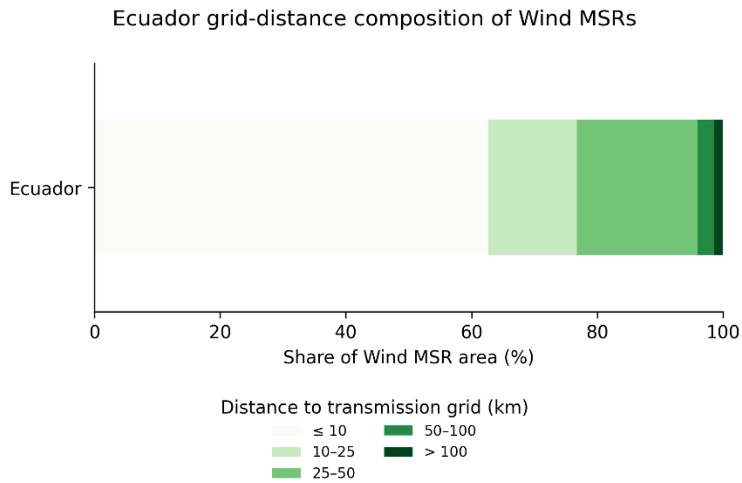

Figure 21: *Example composition of wind MSRs by distance to transmission grid (km) in Ecuador.*

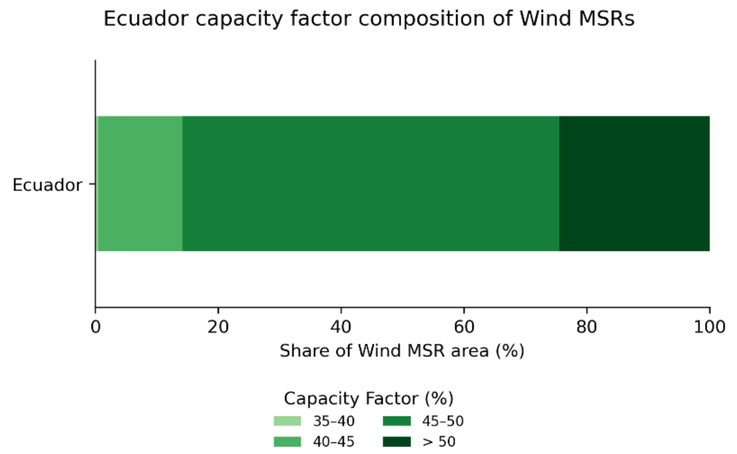

Figure 22: *Example composition of wind MSRs by capacity factor (%) in Ecuador.*

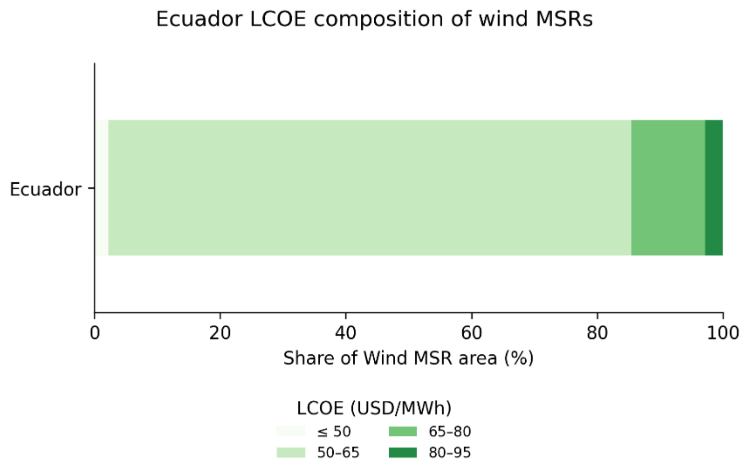

Figure 23: Composition of wind MSRs by LCOE (USD/MWh) in Ecuador.

👉🏼 Go back to the ***control file*** → *configurations* worksheet, and enable solar by setting the *run_code_for_solar* configuration equal to 1 and disabling the other technologies.

🏃 **CODE RUN** 🏃

Run ***attributor_combiner.py*** for solar PV.

The summary figures show the composition of suitable solar PV MSR areas by distance from the grid (example in **Figure 24**), capacity factor (example in **Figure 25**) and LCOE (example in **Figure 26**). We can also investigate how the attributes compositions is distributed spatially across the country by visualizing in QGIS the shapefile created by the Attributor Combiner (**Figure 27**).

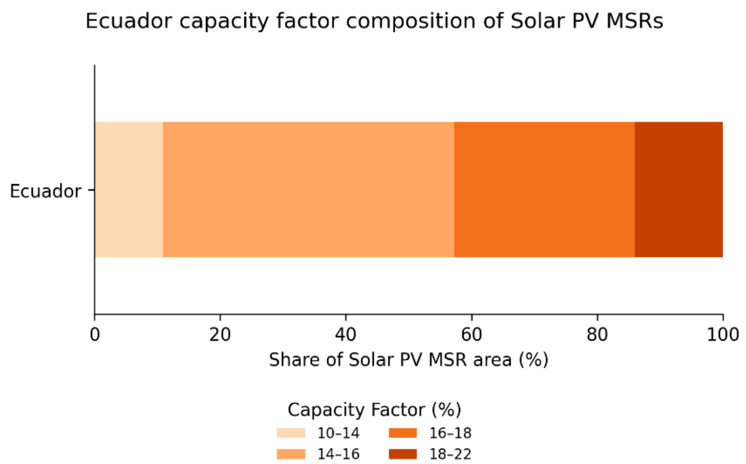

Figure 24: Example composition of solar PV MSRs by distance to transmission grid (km) in Ecuador.

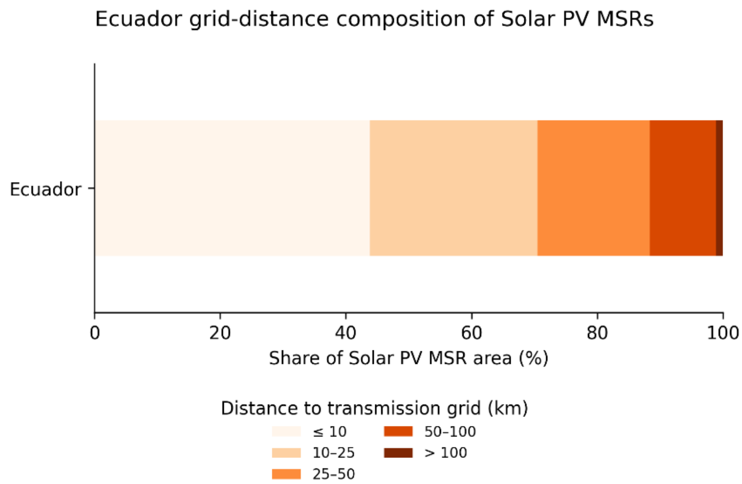

Figure 25: *Example composition of solar PV MSRs by capacity factor (%) in Ecuador.*

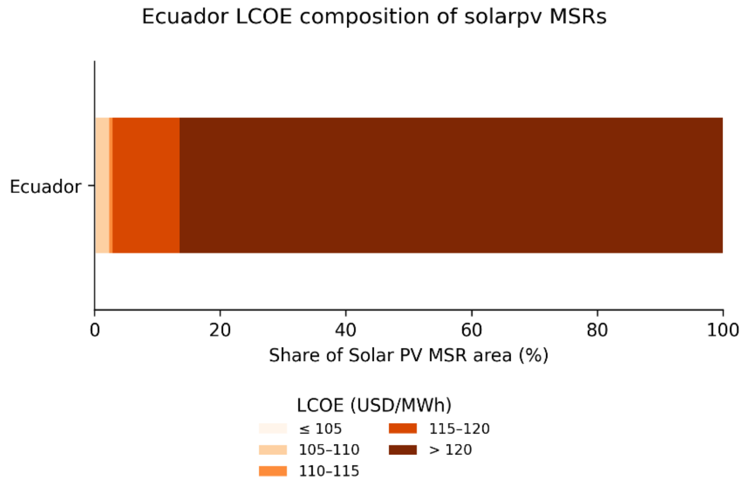

Figure 26: *Example composition of solar PV MSRs by LCOE (USD/MWh) in Ecuador.*

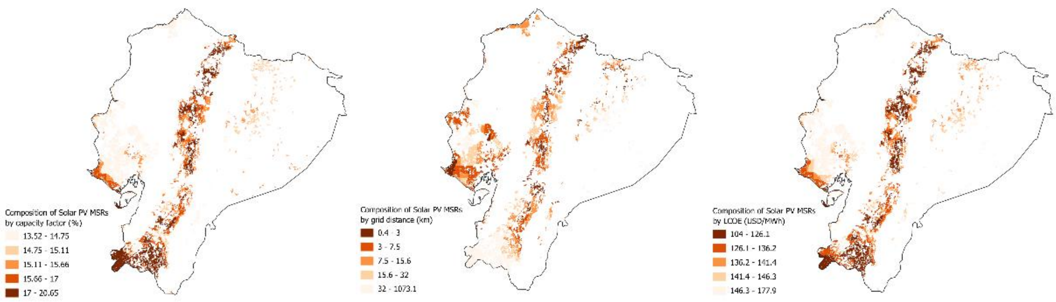

Figure 27: *Example spatial distribution of a) grid distance (km), b) capacity factor (%) and c) LCOE (USD/MWh) composition of solar PV MSRs in Ecuador.*

Part 4: Screener

The next step consists of creating a subset of MSRs suitable for model inclusion. The standard approach in the MSR workflow is to rank the MSRs by expected LCOE (including costs of infrastructure for generation, transmission network extension, and road network extension), and create a sub-selection of those where power generation would be cheapest according to this LCOE.

This sub-selection is put in practice through a user-defined area parameter that allows selecting the best MSRs up to a coverage of a certain percentage of a country's surface area.

Part 4.1: Setting up the control file

Open the **control file** for the Screener script, ***control_file_screener.xlsx***, that includes the same worksheet structure as the other control files.

Open the ***control file*** → *paths* worksheet.

Specify the paths to the ***inputs*** and ***outputs*** folders in the appropriate cells, as in **Figure 5**.

Go to the *datasets* worksheet.

Enter the names of the datasets to be imported, as in **Figure 28**. In particular, make sure the .csv files holding the area cutoff percentages in the ***inputs*** folder are correctly linked.

Have a look at those files as well, and **make sure a 5% cutoff has been allocated to Ecuador**.

	A	B	C
1	datasets	value	comments
2	region_boundaries	Ecuador.shp	Name of region boundaries shapefile (input folder)
3	regions	region_names.csv	Name of region names table (input folder)
4	solarpv_region_area_cutoff_perc	solarpv_region_area_cutoff_perc.csv	Name of regional area cutoff thresholds (input folder) for solar PV
5	wind_region_area_cutoff_perc	wind_region_area_cutoff_perc.csv	Name of regional area cutoff thresholds (input folder) for wind
6			

paths | datasets | configurations | parameters | +

Figure 28*: Linking the correct input datasets in the Sceener* ***control file*** *→ datasets worksheet.*

Go to the *configurations* worksheet.

Set *run_code_for_solar_pv* = 1 and *run_code_for_wind* = 1 (this code can run both simultaneously), as in **Figure 29**. Also set *add_cf_profiles* = 1. This way, the code will provide model-ready hourly time series output of solar and wind capacity factors across the analysed meteorological years (here 2012-2014).

	A	B	C
1	parameter	val	comments
2	run_code_for_solar_pv	1	Options=[0-No, 1-Yes].
3	run_code_for_solar_csp	0	Options=[0-No, 1-Yes].
4	run_code_for_wind	1	Options=[0-No, 1-Yes].
5	run_code_for_offshore_wind	0	Options=[0-No, 1-Yes].
6	local_time_profile	1	Options=[0-UTC time profile, 1-Local time profile].
7	add_cf_profiles	1	Options=[0-No, 1-Yes].
8	run_lcoe_screening_method		Options=[0-No, 1-Yes]. *Not yet operational - for future use*
9	run_variability_screening_method		Options=[0-No, 1-Yes]. *Not yet operational - for future use*
10			

paths | datasets | configurations | parameters | +

Figure 29*: Setting the configuration for the Screener output in the Sceener* ***control file*** *→ configurations worksheet.*

Part 4.2: Running the Screener

> 🏃 **CODE RUN** 🏃
>
> Run *screener.py*.

Follow the log messages in the terminal to monitor the progress of the workflow. The script reports the active region, technology and workflow stage. The run is complete when the message "*Screener workflow completed*" appears in the terminal.

In the ***output*** folder, the script creates a folder *4_screener* with a subfolder for each modelled region. Here, the outputs are stored in the ***Ecuador*** subfolder, where the following files are created:

- Shapefiles for the filtered "best ranked" MSRs for solar and wind power;
- A .csv file with the "best ranked" MSRs and their attributes, including time series.

👉 Have a look at the shapefiles. **Where are the "best ranked" MSRs located spatially**, when compared to the full set of MSRs (output of the MSR Creator)? How can this be explained taking into account **resource strength** and **remoteness metrics**?

👉 Have a look at the output time series in the produced .csv files. The best ranked MSRs are shown in columns, as in **Figure 30**. Scrolling down, one finds (from row 68 onwards) the **full hourly time series of capacity factors** across the analysed period.

	A	B	C	D	E	F	G	
1	MSR_ID	10	17	13	16	14	11	
56	GHIkWhm2d	5.692785	5.692785	5.692785	5.692785	5.692785	5.692785	5.
57	RawERAmean	14.6585	16.46174	14.62807	15.91771	13.30183	14.62807	13
58	CorAdderWh	62.70746	6.281074	58.08763	21.96128	92.56599	52.44209	86
59	CF	20.08435	20.01165	19.98582	20.02749	19.91211	19.72499	19
60	Y_GWh	1079.548	40.01495	518.0958	167.8144	46.9259	100.013	85
61	sLCOE-MWh	101.6887	102.0426	102.1689	101.9653	102.5313	103.4634	10
62	tLCOE-MWh	4.70354	4.596491	4.62463	5.029801	4.657074	4.56418	4.
63	tCAPEX-kW	80.92512	78.79707	79.17716	86.2935	79.43854	77.12236	81
64	rLCOE-MWh	0.272422	0.107149	0.282454	0.136528	0.183942	0.211703	0.
65	rCAPEX-kW	4.350587	1.704982	4.488673	2.174185	2.912368	3.320411	2.
66	LCOE-MWh	106.6647	106.7462	107.076	107.1316	107.3723	108.2392	10
67	trCAPEX-kW	85.27571	80.50205	83.66583	88.46768	82.3509	80.44277	83
68	1-1-2012 00:00	0	0	0	0	0	0	
69	1-1-2012 01:00	0	0	0	0	0	0	
70	1-1-2012 02:00	0	0	0	0	0	0	
71	1-1-2012 03:00	0	0	0	0	0	0	
72	1-1-2012 04:00	0	0	0	0	0	0	
73	1-1-2012 05:00	0	0	0	0	0	0	
74	1-1-2012 06:00	0	0	0	0	0	0	
75	1-1-2012 07:00	0.084	0.031	0.081	0.05	0.107	0.075	
76	1-1-2012 08:00	0.254	0.196	0.275	0.231	0.258	0.269	
77	1-1-2012 09:00	0.479	0.518	0.521	0.524	0.485	0.516	
78	1-1-2012 10:00	0.645	0.689	0.704	0.692	0.694	0.699	
79	1-1-2012 11:00	0.733	0.824	0.698	0.804	0.598	0.694	
80	1-1-2012 12:00	0.81	0.819	0.821	0.783	0.75	0.816	
81	1-1-2012 13:00	0.728	0.787	0.69	0.76	0.742	0.686	
82	1-1-2012 14:00	0.627	0.694	0.615	0.684	0.673	0.61	

solarpv_5.0%_screened_msrs

***Figure 30**: Full hourly time profiles for the "best ranked" MSRs (example for solar PV).*

Part 5: Clustering

[not yet completed]